

Weekly Report Generator & Team Dashboard

TECHNICAL ASSESSMENT

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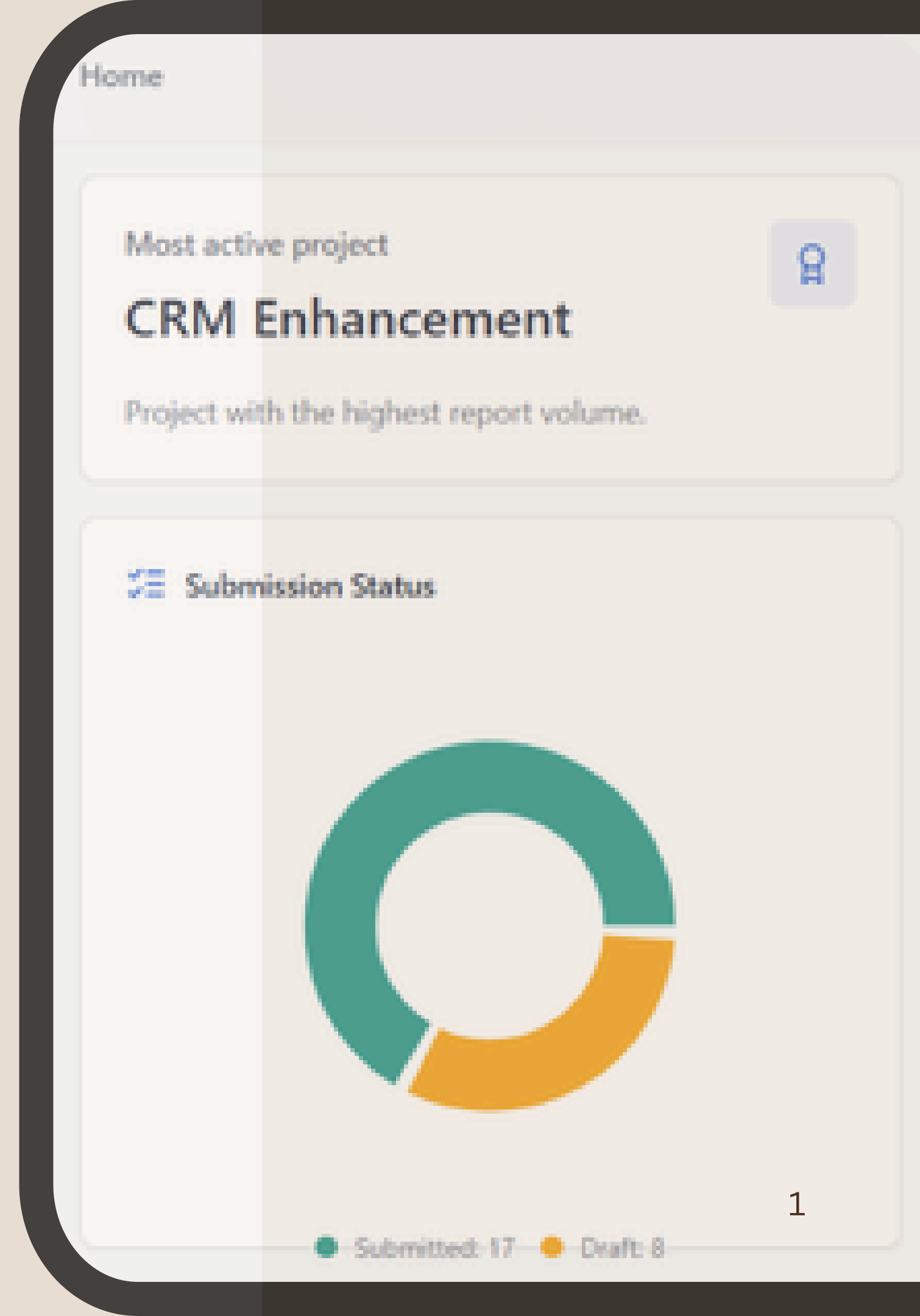


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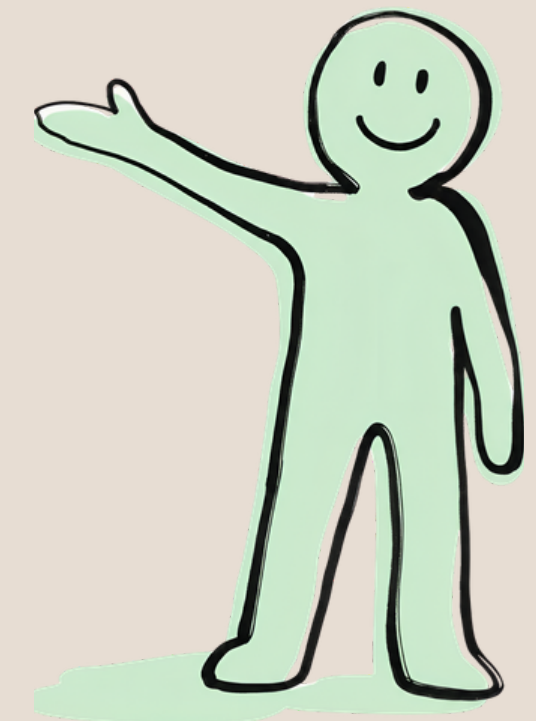
Introduction

Project Overview

The Weekly Report Generator & Team Dashboard is a web-based system that helps organizations manage projects and streamline weekly reporting through a centralized platform.

Purpose

- Simplify weekly report submission.
- Improve project monitoring.
- Provide real-time dashboard insights.
- Support data-driven decision making.





Problem Statement

- Many organizations still rely on manual methods such as spreadsheets or emails for weekly reporting.
- Managers face difficulties in tracking project progress and monitoring team performance.
- Delayed or incomplete report submissions reduce productivity and decision-making efficiency.
- There is no centralized platform to manage projects, reports, and analytics in one place.
- Lack of real-time insights makes it difficult to identify project risks and reporting compliance.



Solution

- Develop a centralized web-based Weekly Report Generator & Team Dashboard.
- Implement secure Role-Based Access Control (RBAC) for managers and team members.
- Enable structured weekly report creation, editing, and submission.
- Provide real-time dashboard analytics for project and team performance.
- Integrate an AI Assistant (Ollama) to generate reporting insights and answer project-related queries.
- Store all users, projects, and reports in a centralized PostgreSQL database.

Project Objectives

-  Develop a secure web-based weekly reporting system.
-  Simplify project and weekly report management.
-  Improve team collaboration and productivity.
-  Provide real-time dashboard analytics and reporting insights.
-  Implement Role-Based Access Control (RBAC) for secure access.



Technology Stack

Frontend

- React.js
- Vite
- TypeScript
- Tailwind CSS

Backend

- Node.js
- Express.js
- Prisma ORM

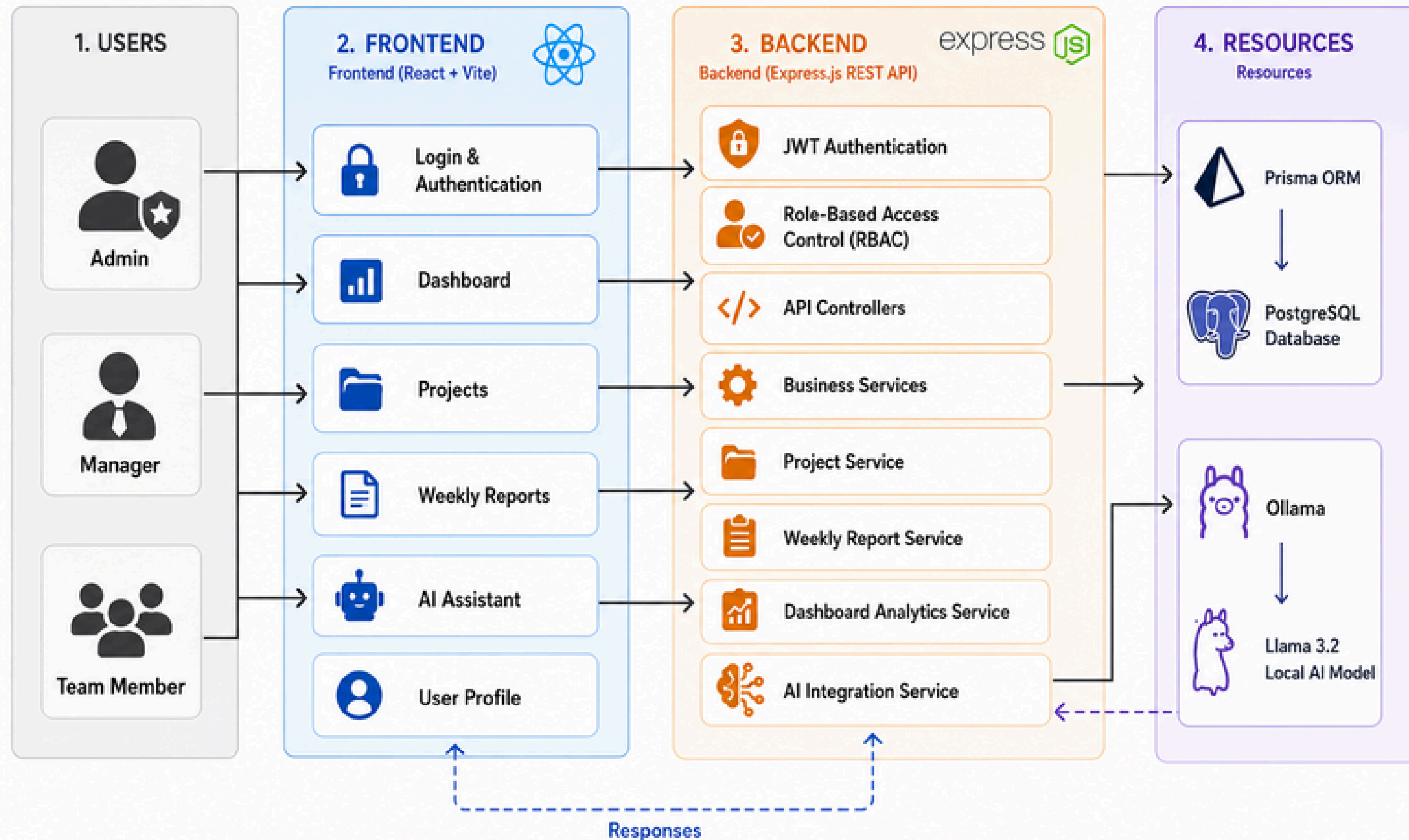
Database

- PostgreSQL
- Authentication
- JWT
- Bcrypt

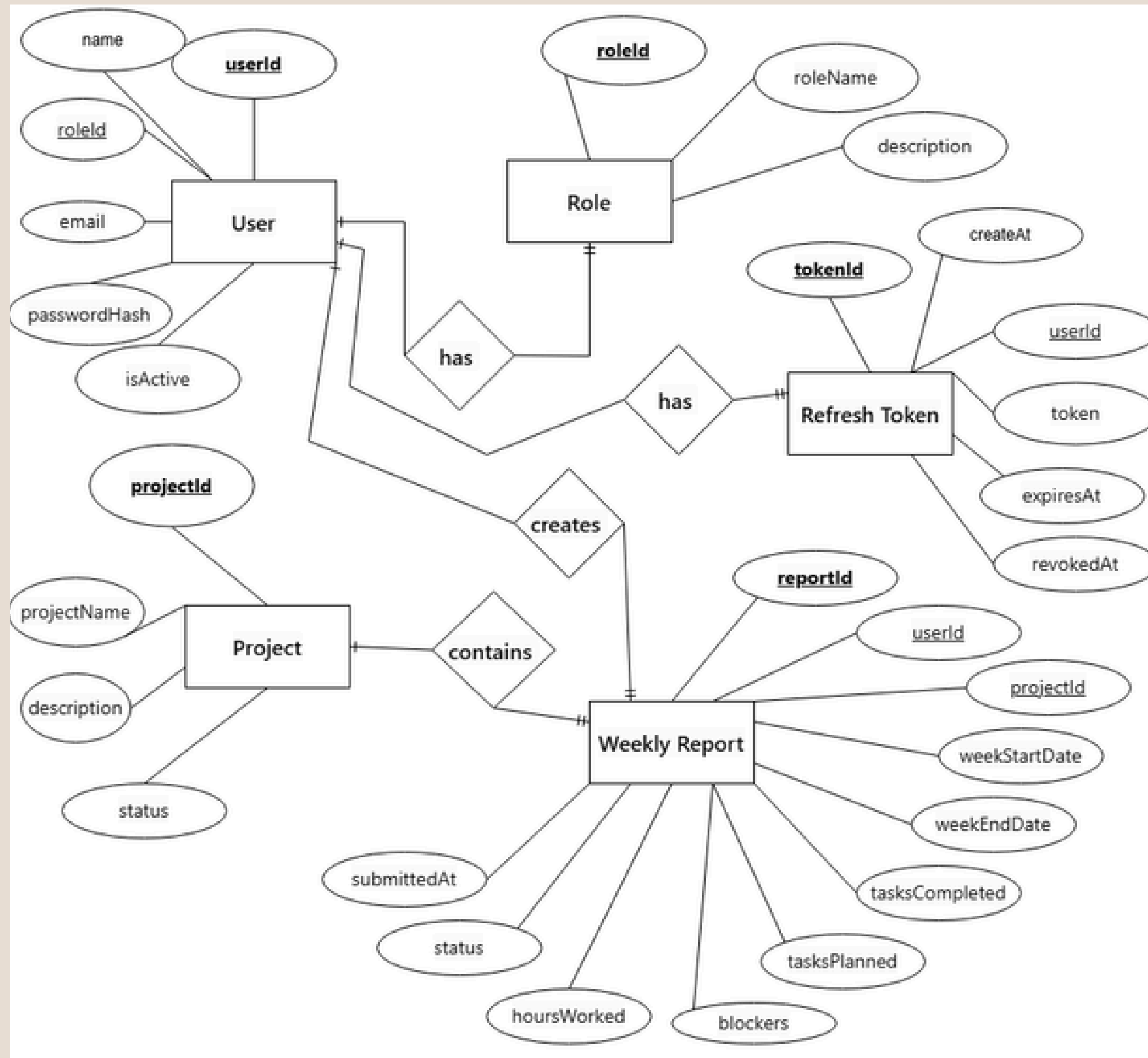
AI Integration

- Ollama
- Llama 3.2

System Architecture Overview



Database Design (ER Diagram)



API Design & Backend Structure

 RESTful API

 Controllers & Services

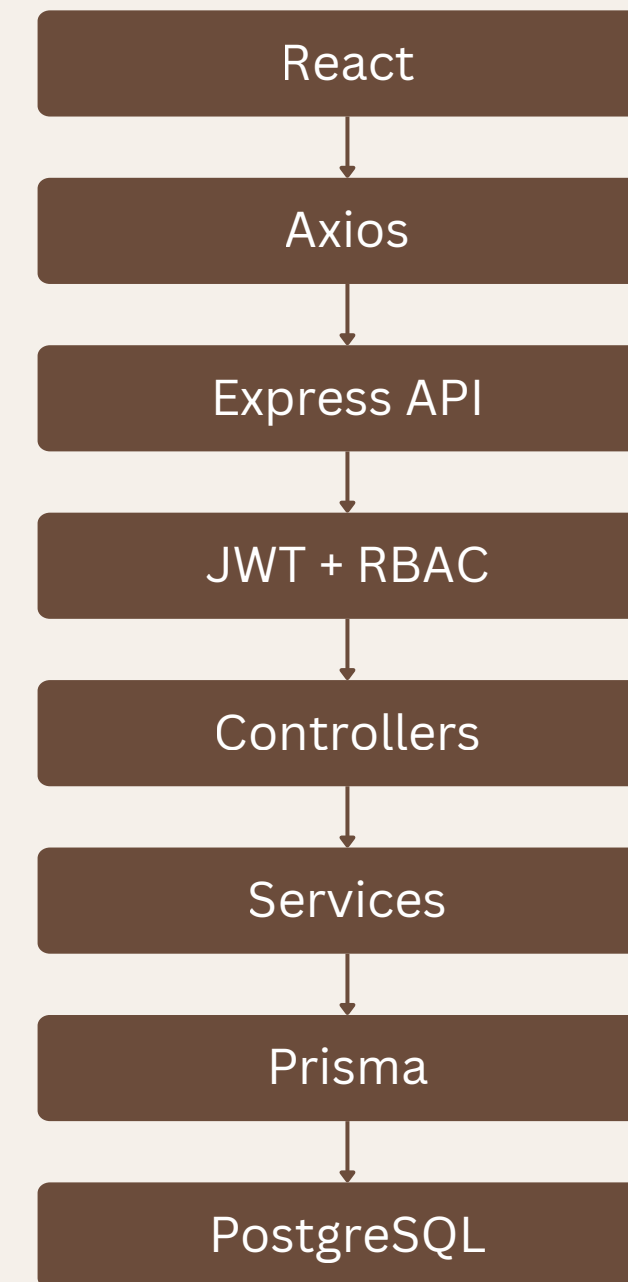
 Prisma ORM

 JWT Authentication

 Role-Based Access Control

 Zod Validation

Request Lifecycle



Core Features

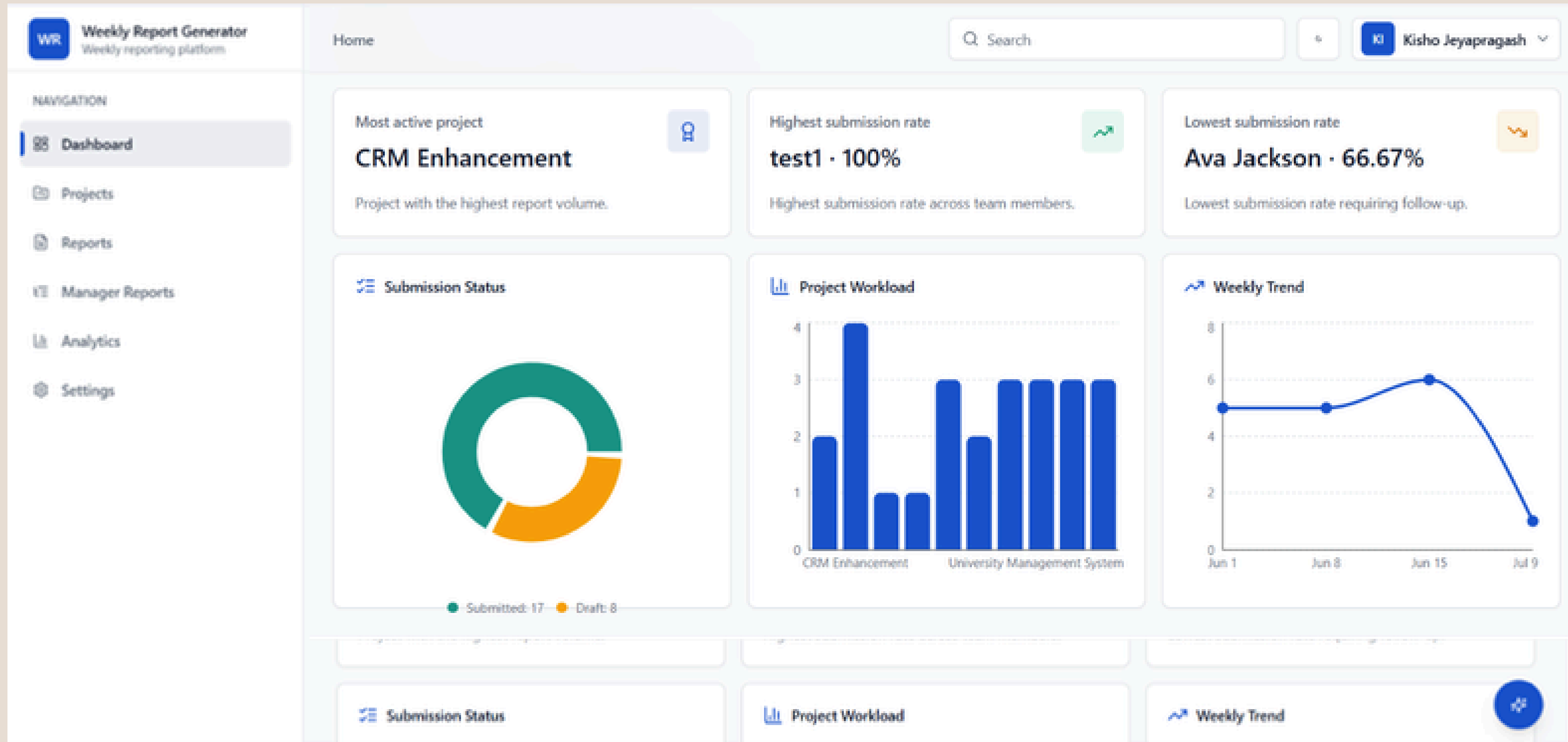


- User Authentication & Role-Based Access Control (RBAC)
- Project Management (Create, Edit, Archive Projects)
- Weekly Report Management (Create, Update, Submit Reports)
- Interactive Dashboard & Analytics
- AI-Powered Assistant (Ollama – Llama 3.2)
- Responsive & Secure Web Application

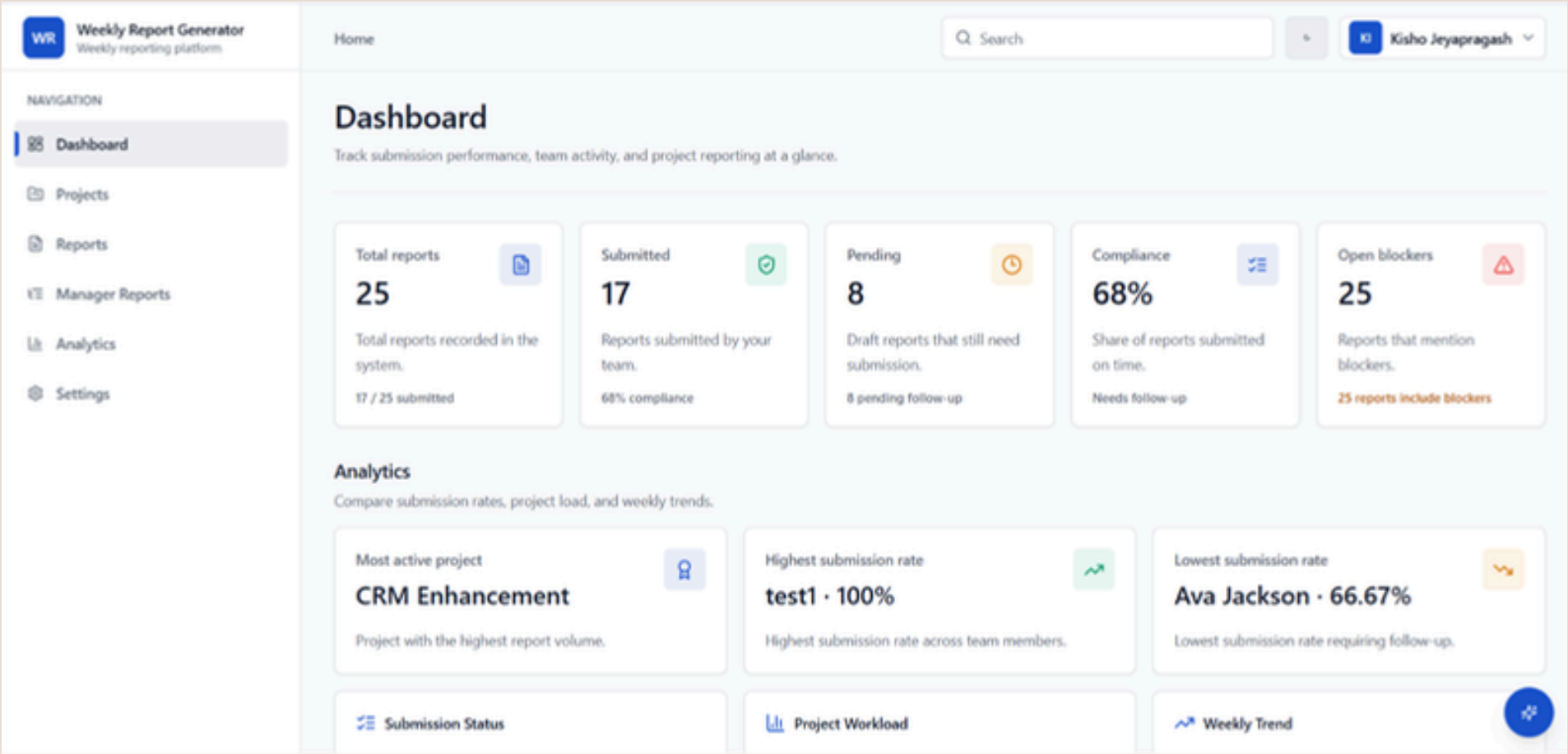
Project Modules

- Authentication Module – User login, registration, and role-based access control.
- Project Management Module – Create, update, view, and archive projects.
- Weekly Report Module – Create, edit, submit, and manage weekly reports.
- Dashboard Module – Monitor project progress, reporting status, and analytics.
- AI Assistant Module – Provide intelligent insights and answer project-related queries using Ollama.
- Profile & Settings Module – Manage user profile, account settings, and preferences.

Dashboard Analytics



Team Member Dashboard and Manager Dashboard



Weekly Report Generator
Weekly reporting platform

Home > Reports

Search

Liam Anderson

Weekly Reports

Create, review, and submit weekly reports.

Search reports

All statuses

All projects

mm/dd/yyyy

Newest

+ Create report

WEEK	PROJECT	STATUS	HOURS WORKED	CREATED	ACTIONS
Jun 15, 2026 - Jun 21, 2026	Mobile Banking App ACTIVE	Submitted	44	Jul 8, 2026	+ / < > ⋮
Jun 8, 2026 - Jun 14, 2026	HR Management System ACTIVE	Draft	39	Jul 8, 2026	+ / < > ⋮
Jun 1, 2026 - Jun 7, 2026	Customer Portal Revamp ON_HOLD	Submitted	34	Jul 8, 2026	+ / < > ⋮

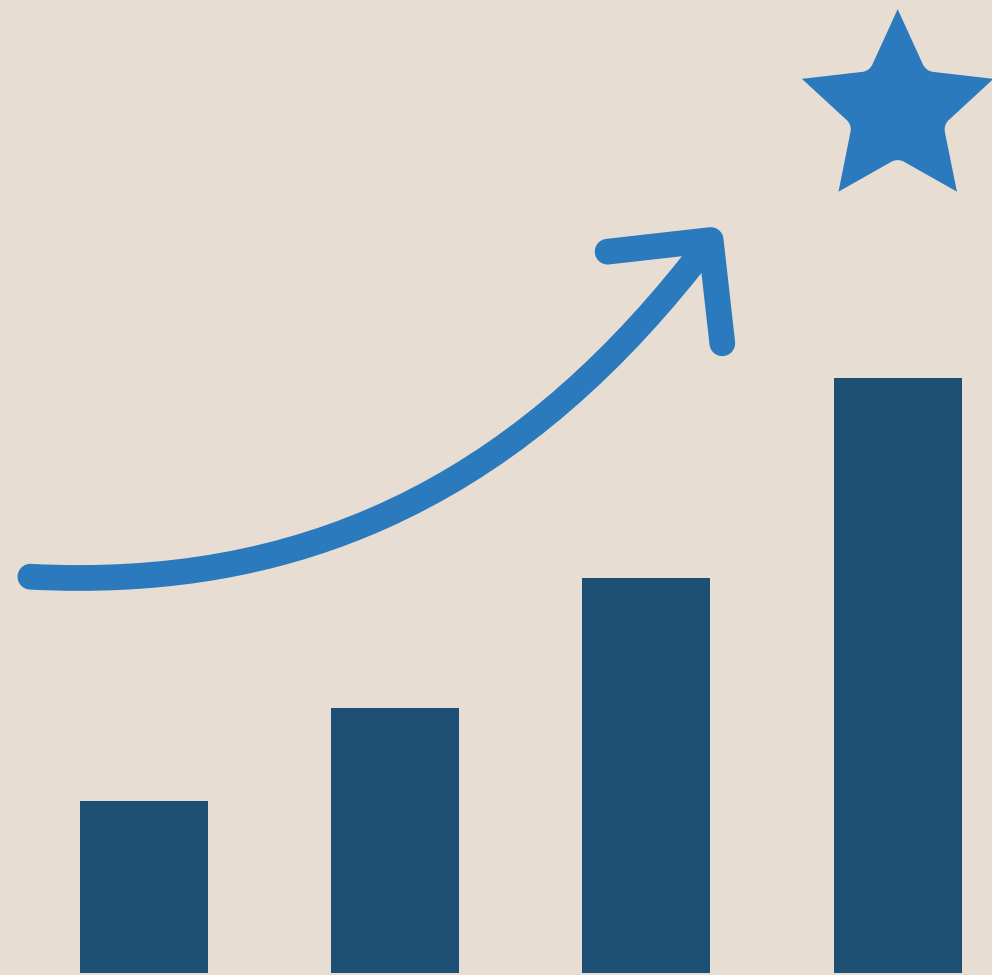
AI Assistant

AI Assistant Features

- Provides intelligent responses to project and reporting queries.
- Summarizes weekly reports and project progress.
- Identifies pending tasks and reporting status.
- Generates quick insights to support managerial decisions.
- Powered by Ollama (Llama 3.2) for secure local AI processing.



Future Improvements



- Implement Email Notifications for report reminders and updates.
- Add PDF and Excel Export for weekly reports.
- Develop a Mobile Application for Android and iOS.
- Integrate Cloud Deployment and CI/CD pipelines.
- Support File Attachments in weekly reports.
- Enhance AI capabilities with advanced analytics and predictive insights.

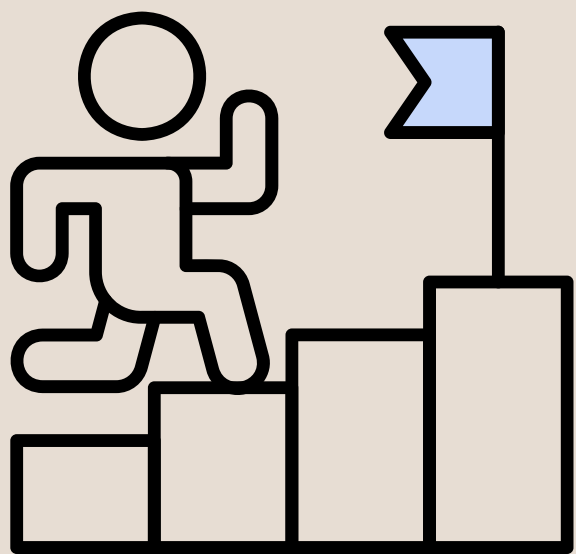
Challenges & Solutions

Challenges

- Secure user authentication
- Managing project and report data efficiently
- Tracking team progress in real time
- Ensuring data validation and security
- Integrating AI into the reporting system

Solutions

- Implemented JWT Authentication with Role-Based Access Control (RBAC).
- Designed a relational PostgreSQL database using Prisma ORM.
- Developed an interactive dashboard with analytics and reporting insights.
- Applied Zod validation, bcrypt password hashing, and protected REST APIs.
- Integrated Ollama (Llama 3.2) to provide AI-powered insights and assistance.



Conclusion

- Successfully developed a secure and scalable Weekly Report Generator & Team Dashboard.
- Simplified project management and weekly reporting through a centralized platform.
- Implemented Role-Based Access Control (RBAC) and secure JWT authentication.
- Delivered real-time dashboard analytics for effective project monitoring.
- Integrated an AI Assistant (Ollama – Llama 3.2) to provide intelligent reporting insights.
- Built using modern technologies to ensure maintainability, performance, and scalability.



Thank You!

Q/A?